

Radiometric characterization of *Erythrina verna* Vell.: A study of natural radioactivity in plant tissues.

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Introduction

The radiometric characterization of native and/or endemic plants from tropical climates represents a knowledge gap identified since the data published in 2003 by the International Atomic Energy Agency (IAEA) in ICRP Publication 108, which established Reference Animals and Plants (RAPs) and emphasized the need for information on non-human biota. The species *Erythrina verna* is native to Brazil, although described as endemic to Brazil and native to the southeastern region, occurrences have also been reported in Bolivia and Peru.

Considering this context, and the broad biodiversity of the Atlantic Forest biome, the aim of this study was to evaluate the radiometric profile of *Erythrina verna* Vell., popularly known as mulungu.



Erythrina verna Vell. By: Martins, M. V. 2020

Methodology

The Specimen collection was carried out at the Botanical Collection Sítio E-Jardim, located in the municipality of Duque de Caxias (22°34'14.6"S; 43°16'17.3"W). Samples were obtained from a cultivated individual, including stem, leaves, and the soil surrounding the roots. The stem and leaves were cleaned to remove any residues, ensuring that only vegetative structures were preserved for analysis. Each plant structure and the soil were stored separately in polypropylene containers and kept at -4 °C for at least 28 days to allow secular equilibrium to be reached.

The samples were analyzed using a High-Purity Germanium (HPGe) detector for 28,800 s to obtain emission spectra and quantify ²²⁶Ra, ²²⁸Ra, and ⁴⁰K. The ²²⁶Ra concentration was quantified using the ²¹⁴Bi peak at 609.3 keV, ²²⁸Ra using the ²²⁸Ac peak at 969.1 keV, and ⁴⁰K using the peak at 1460.8 keV.

Results

The results demonstrate that no significant absorption process occurred for the analyzed radionuclides, as only small amounts of ²²⁶Ra, ²²⁸Ra, and ⁴⁰K were detected in the vegetative structures compared to the concentrations present in the soil. This observation may be associated with the physiological characteristics of nutrient uptake and osmoregulation in the plant.

Table 1 - Activity Concentration in Bq.kg⁻¹.

Sample	²²⁶ Ra	²²⁸ Ra	⁴⁰ K
Soil	26,50	49,50	133,54
Stem	0,74	0,63	17,34
Leaf	0,14	1,58	22,38

When assessing soil-to-plant transfer for the analyzed radionuclides, ⁴⁰K showed transfer rates of 13% to the stem and 17% to the leaves, totaling 30% for the whole plant. These values are considered low when compared with those reported for other species cited in the literature and commonly used as food (Table 2). For ²²⁶Ra and ²²⁸Ra, transfer values ranged from 1% to 3%, totaling 3%–4%. It is important to note that roots were not analyzed; therefore, the values reported in this study are likely underestimated.

Samples	Transfer-factor of Radionuclides (whole plants)			References
	²²⁶ Ra	²²⁸ Ra	⁴⁰ K	
Chinese cabbage	~ 0.100	-	-	Smodiš et al., 2015
Cucumber plant	2.50	0.80	6.00	
Sweet pepper plant	-	-	7.50	Hegazi et al., 2023
Hot pepper plant	0.40	0.60	6.70	
Tomato plant	0.30	-	8.60	
<i>Apium graveolens</i> from Baghdad	-	0.13	1.76	Mostafa et al., 2021
<i>Raphanus sativus</i> from Baghdad	-	0.19	2.93	
<i>Ocimum basilicum</i> from Baghdad	-	0.26	2.18	
<i>Erythrina verna</i>	0.03	0.04	0.30	Current study

Conclusion

Thus, it can be preliminarily inferred that teas, infusions, and other preparations derived from *Erythrina verna* do not exhibit an accumulation profile for the radionuclides Ra and K. Furthermore, the soil at the sampling site showed ⁴⁰K concentrations below the reference levels established by UNSCEAR and those reported for other regions of southeastern Brazil.

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References

